

Company: MSCI

Job Profile: Technology and Data Operations Intern

Offer Type: Internship with Performance based PPO

Internship Stipend: 60K

CTC: 14.68 LPA (Base 13.2 LPA)

Location: Mumbai/Pune

Process Date: OA on 11th July, Interviews on 15th July 2025

Placement Process:

1. OA: (75 minutes) (~240 Candidates appeared)

This was an online assessment consisting of three questions: 2 DSA based and 1 SQL, on Hackerrank. The DSA questions were randomly selected from a pool of questions.

- DSAQ1: (<https://www.geeksforgeeks.org/dsa/python-program-to-find-uncommon-words-from-two-strings/>) The first question was to **find uncommon words from two strings**, given the full sentence string separately. This was an easy question using `split()` and a two-pointer approach.
- DSA Q2: (<https://leetcode.com/problems/maximum-OR/description/>) The second question involved maximizing the OR value (OR of all elements) in an array, given you can multiply any element by 2, at most `k` number of times. I did manage to pass all test cases for this one, although not with an optimal solution.
- SQL Q3: This was a SQL question involving two tables related to voter information.

The goal was to return a summary of each state, showing the names of voters who voted in that state along

with the number of times each voter voted.

Given Tables:

Voters: Votes:

voter_id | name voter_id | state

-----|----- -----|-----

1 | Tanvi 1 | California

2 | Gaurang 2 | California

3 | Drishya 2 | California

4 | Rahul 1 | California

4 | Texas

Expected Output :

state | voters_summary

-----|-----

California | Tanvi x 2, Gaurang x 2

Texas | Rahul x 1

I wasn't able to solve this question, as it involved use of CTEs and Group_Concat, etc..

Solving 2 DSA questions fully was enough to qualify for the next round, which I did. **Around 41 students were shortlisted for Tech Round 1.**

2. Offline Technical Round 1: (30-45 minutes)

I was **firstly asked to introduce myself**. Then, the interviewer delved into my resume, **particularly focusing on a GenAI project**, which I described to him in detail. He further questioned me about **how vector databases work in LangChain, and what cosine similarity** is. I was well aware of these topics and answered them appropriately.

He also asked me about **static and dynamic components in React**. Since I'm not primarily a frontend person, I mentioned the single basic React project I made in my first year, how I used components as reusable entities, and what I could make out by the name dynamic components, while clearly mentioning that it had been a while for me working with React and frontend in general.

He also asked me **what I knew about MSCI** – I had researched the website beforehand and gave the appropriate answer.

Then, I was asked a **simple question of reversing a string, using OOP**. I just created a class with a string as an attribute, and a method for string reversal. And the interviewer was satisfied with that.

A **SQL query was presented**, based on **INNER**, **LEFT**, and **RIGHT JOIN** where tables contained equal entries (These were two different tables with only 'id' as the attribute). This was more of a trick question as all these joins would result in the same table with 6 rows where **a.id = 1** and **b.id = 1**. I too found it tricky first and described the theory about these joins, stating that, according to my knowledge, all the tables should be the same, as mentioned above. To which the interviewer replied that yes, this is the right answer.

Another SQL problem involved: Given a table **Products (pid, ..)** and **Orders (pid, oid)**, find the products that are unsold. The key here was to consider that orders are only logged for products that are sold (as there is no 'status' column). This could then be solved by simply **LEFT JOINING Products and Orders** and selecting where **oid = NULL**.

A puzzle was given: 6 balls of different weights and a weighing machine. Find the heaviest ball while using the weighing machine the minimum number of times. Here I was unable to arrive at a number less than 6 with various combinations. So he gave me a hint that we can also put multiple balls on the scale at once. Then I realized we can split the balls into groups of 3 and pick the heavier group and repeat a similar process with the heavy group.

Then he opened for questions. First, I asked him about himself. Then I asked a question about MSCI particularly, that given they manage such huge amounts of indexes, there must be a large amount of data to be processed, and that too in real time, so how is all this managed. His answers to both the questions were very detailed and he seemed happy.

A proper shortlist wasn't announced in this round, as after the first round itself, people who were selected were called for the second, while those who were rejected were informed so. **Around 20-22 students may have been selected in total.**

3. Offline Technical Round 2: (45-60 minutes)

"How does your typical day go?" – This was a question that I wasn't expecting or prepared for as the first question. I answered covering my college routine, assignments, LeetCode practice, and working on projects/hacks.

"What have you done in your college life overall?" – I answered this honestly, explaining that I explored React in my first year but realized it wasn't my cup of tea, then started with Django and also learning ML since last one year, also mentioning my elective "build LLM from scratch" which I had taken last sem as a PE.

"What are your hobbies?" – I play chess as a hobby, my rating is 1000 rapid on chess.com and around 700 in blitz and bullet. Apart from that I also play keyboard as a hobby, again not professionally, but as a hobby.

"Write down an algorithm to teach someone new to the keyboard." – The question was ambiguous, so I asked for clarification. To which he said, "just describe how you would teach keyboard to a beginner." So I explained the entire process from basic keys, chords, and once they reach a certain level (which can be determined by tests), teaching them basic songs, and advanced songs once they advance tests.

"Design an ER diagram based on whatever I explained" - I started with a users table, a courses table, a courseTaken table (with FK to user_id and course_id) which can be joined, then a table for lessons (FK to course_id), and tests (FK to lessons). The key here is not to describe each and every table or attribute, but to describe the core tables and keep them normalized, avoiding any redundant tables, etc. Since I have designed SQL schemas for a lot of my Django projects, I was fairly confident in designing and explaining this and also ready to explain further.

Then I was asked about my projects which I started to describe. The interviewer interrupted and asked about a particular project, which was the Alumni-Connect App I made during the SIH finals. I described all the features correctly. He then asked **how the alumni would benefit from this**, to which I replied we also had a job portal where alumni can list referrals they have, which would then also benefit them. Also, our college has a business incubation system where student-startups can be incubated and they can also hire students from the college through this portal. Then he **asked about the role of AI in this project as my resume and intro pointed towards that**. I said that in this project there was a very small role of AI, in the FAISS based recommendation system to recommend alumni/student profiles based on a user's own profile/resume. I explained the working in detail.

Then he asked about my profile, being focused towards AI and Python, **what I would do if I knew there was remotely no use of AI and Python in my role**. I answered that, given enough time, I would learn whatever technology is offered, while also making suggestions from my side that maybe AI can make this particular part of the application better, and maybe we can integrate an AI part externally through APIs, while not moving away from the main tech stack of the applications, may it not be Python.

Puzzle: "Split or Steal" – Two people play a game, each time: if they both split, they get equal reward; if one splits and one steals, the stealer gets all of the reward; if both steal, no one gets the reward. This is repeated for n turns (n=20,100). What strategy would you use to maximize your reward? This is more to check the thought process rather than a right answer. I too started with the simpler strategy to always steal, so we at least win the game, even with minimal reward, but then the interviewer told me, "why do you think the other will also always steal?" To which I gave the better solution: to start by splitting, and if the other person steals repeatedly, we too start to steal a couple of times, to indicate that "We're not giving steals for free" and the other person would also not earn anything if they kept stealing.

He then asked me about **GROUP BY and HAVING in SQL**. I did answer that it is used to filter after the use of GROUP BY, within the groups, but I think I could have maintained more clarity in that answer. He then asked **me the difference between data warehouses and**

relational databases. I didn't have much idea of warehouses other than the fact that they're used to store large amounts of structured data, while relational databases are used in order to describe relations and whenever there are joins. He did grill me that joins can also be done in warehouses, to which I agreed that I didn't have much idea about warehouses. I told that relational databases are particularly useful to show relations like foreign keys, example: posts, comments, etc.

Then he opened for questions, and I asked him about himself (**because MSCI interviewers generally don't introduce themselves at the start and the HR may ask who you interviewed with to check you**). He was an SPIT alum himself, and worked at MSCI on the data engineering side. Then I asked about what he expected from freshers at MSCI to which he replied smilingly that "I don't expect much from freshers except basic logical and problem-solving skills and decent projects, and once they join in they can be groomed according to the role if the basic requirements are considered."

Around 10 students were selected for this round.

4. Offline HR Round: (30 minutes)

The HR Round was very chill and I was asked very basic questions:

"Introduce yourself professionally," based on my projects and as a person, and where I live. My branch is Data Science, but my projects were ML-based, so I was asked about the **difference between ML and Data Science and what I was actually interested in,** to which I replied that ML and Data Science are dependent upon each other, and that I'm trying to explore the entire pipeline, and ML is just a part which helps overall.

She asked about my **family background.** She too asked, **"Do you know about MSCI?"** and **"How do you know about it?"** – I told her whatever I researched and that I knew about MSCI since it has been coming to college for placements for the last few years. **She asked if I invest or plan to invest** – I don't invest currently, as I am less into the finance side, but I plan to invest.

"Are you planning on any further education like masters/MBA?" – If I were to opt, I would choose MS over an MBA because of my interest in tech. But still, I find many courses from top institutions online itself, which makes me question whether doing a masters is worth it. Hence, I want to first work in the industry and while focusing on upskilling myself through such online courses and then take a call on it. She also asked whether I would be able to relocate / which location I would prefer.

She opened for questions. I asked a couple of questions, starting with what was being expected from me, to which the HR replied that the main thing they expect is clarity of thought regarding future goals. (She mentioned that people who are considering masters straightaway may lack focus in the organization and must at least try to stay 3-4 years before taking a call). My previous answers were very well aligned to her expectations. Secondly, I just asked for general feedback (you can also ask this from tech interviewers). To which she

replied that I was moving my chair throughout, which showcases my nervousness, but that she understood that this may happen during an interview and advised not to do it generally.

5 students were selected.

Tips:

- Even if you are not super good at DSA, do practice the standard problems from **Striver's SDE sheet, or keep doing LeetCode daily**, so that you are able to solve at least medium LeetCode questions comfortably. I also **kept giving contests on codeforces and kept a timer on LeetCode problems for timed prep**. Hard questions are generally rare beyond the very top companies like PhonePe.
- Also, practicing and completing the **LeetCode SQL 50** a week before or even during the placement would give decent experience of SQL.
- For System Design, I referred multiple sources. **Piyush Garg** (<https://www.youtube.com/@piyushgargdev>) and **Arpit Bhayani** (<https://www.youtube.com/@AsliEngineering>) on Youtube have some top quality content. But do explore other videos as well, some creators may explain niche topics better.
- For core subjects too, I did not have a fixed playlist or resource. I explored different videos or GFG articles for different concepts, even clarified some questions in depth through GPT/Gemini as well. Some good resources include **GateSmashers** (<https://www.youtube.com/@GateSmashers>), and again **Arpit Bhayani**. Make sure you know what things are asked commonly, **focus on understanding maximum concepts, through any videos or articles or questioning GPT on the same**.
- **Prepare your introduction and some general HR questions or description about your projects/internships/committee roles** if asked in detail and have it down in written word for word. You can use AI tools to enhance this. You may not be able to speak it exactly how you've written, but it helps you maintain a proper flow each time you encounter a particular question which you've prepared and increases confidence while answering. **Confidence and clarity was an important factor** they checked other than technical knowledge as I mentioned above.
- **Read all the interview experiences of that particular company before interviews, not just 1 or 2.** Questions asked are mostly on similar lines, or you get an idea that what variety of questions are asked. For example, **LC easy to medium level DSA questions were asked to people in other panels**,. Few I could remember are **Next Permutation (DP based) and Left Hand View of a Tree**, as well as **Buy and sell stock questions on LeetCode**.
- I would agree that luck did play some part as I did not get asked any DSA question, which made the process relatively easier for me. But **at the same time, I did mention one-liners about all my projects in my general intro in all rounds, which perhaps helped steer all interviews towards the resume/dev side.** You can also **prepare intros pointing towards your strengths**, may it be DSA, dev or whatever,
- .During these interviews, there were also EDS OAs going on to which I hadn't applied, but some of my friends did, and some of them had to reschedule his interview

slot to a later timing during the OA, which (maybe not by itself) played a role in them not getting selected despite a good first-round interview. **Do not juggle yourselves between two companies, stay focused for one and prepare for it.**

- **Do not be demotivated if you're rejected.** Even I was rejected right at the HR round during my interviews at WorkIndia as only 2 people were hired. I was sad for a day, but then moved on soon, and began my further preps. Companies come till October, that offer decent packages, some even more than this, **move on as soon, and start the grind forward**, If you consider yourself to be at least average level throughout college, **You will get placed sooner or later, with a decent package. Hence do not give up. ALL THE BEST for your placements!**

- Feel Free to connect on LinkedIn if you need any help
<https://www.linkedin.com/in/aroosh-joshi-59601524b/>